

CASE STUDY 1: Banking Loan Processing System

Scenario Question

A bank manages customer loans.

The system must support:

- multiple loans per customer
- reporting and analytics
- safe updates when loan is approved
- data import from CSV/JSON
- structured but flexible data
- relationships between customers and loans

Design and analyze the system using the data below.

Table 1: customers

customerId	name	city	loanIds (array)
C1	Arun	Chennai	[L1, L2]
C2	Meena	Delhi	[L3]
C3	Rahul	Mumbai	[L4]
C4	Kiran	Pune	[L5]
C5	Priya	Chennai	[L6]
C6	Suresh	Trichy	[L7]
C7	Divya	Vellore	[L8]
C8	Ramesh	Karur	[L9]
C9	Anita	Chennai	[L10]
C10	Neha	Madurai	[L11]

Table 2: loans

loanId	customerId	amount	status
L1	C1	200000	Approved
L2	C1	150000	Pending

L3	C2	300000	Approved
L4	C3	180000	Approved
L5	C4	100000	Rejected
L6	C5	250000	Approved
L7	C6	120000	Pending
L8	C7	175000	Approved
L9	C8	90000	Approved
L10	C9	220000	Pending

CASE STUDY 2: Online Course Enrollment System

Scenario Question

An online learning platform manages:

- students
- courses
- enrollments

The system must support:

- many students enrolling in many courses
- flexible student profiles
- reporting for courses and enrollments
- validation of student data
- bulk data import and export

Design and analyze the system using the data below.

Table 1: students

studentId	name	city	courseIds (array)
S1	Arun	Chennai	[C1, C2]
S2	Meena	Delhi	[C2]
S3	Rahul	Mumbai	[C3]
S4	Kiran	Pune	[C1]
S5	Priya	Chennai	[C3]

S6	Suresh	Trichy	[C2, C3]
S7	Divya	Vellore	[C1]
S8	Ramesh	Karur	[C2]
S9	Anita	Chennai	[C1, C3]
S10	Neha	Madurai	[C2]

Table 2: courses

courseId	courseName	fee
C1	MongoDB	5000
C2	NodeJS	4000
C3	Python	4500
C4	React	5500
C5	Java	6000

CASE STUDY 3: Airline Reservation System

Scenario Question

An airline reservation system manages:

- passengers
- flights
- bookings

The system must support:

- multiple passengers per flight
- multiple bookings per passenger
- flexible travel details
- reporting on bookings and flights
- data validation
- safe booking confirmation
- import/export of booking data

Design and analyze the system using the data below.

Table 1: bookings

bookingId	passenger	flightId	seats (array)	status
B1	Arun	F1	[12A]	Confirmed
B2	Meena	F2	[10B]	Pending
B3	Rahul	F1	[14C]	Confirmed
B4	Kiran	F3	[8A]	Cancelled
B5	Priya	F2	[11D]	Confirmed
B6	Suresh	F3	[9B]	Pending
B7	Divya	F1	[15A]	Confirmed
B8	Ramesh	F2	[12C]	Confirmed
B9	Anita	F1	[16B]	Pending
B10	Neha	F3	[7C]	Confirmed

Table 2: flights

flightId	route	price
F1	Chennai – Mumbai	6000
F2	Delhi – Goa	7000
F3	Pune – Jaipur	5000
F4	Chennai – Delhi	6500
F5	Mumbai – Goa	7200